



Research Notes - Anthropic's Global Workspace / J-Space in LLMs

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Research: Anthropic's "Verbalizable Representations Form a Global Workspace in Language Models" (J-space / Jacobian lens)

Date: 2026-07-07 Search queries / sources fetched:

- Announcement:
<https://www.anthropic.com/research/global-workspace>
- Paper:
<https://transformer-circuits.pub/2026/workspace/index.html>
- External commentary PDF (Dehaene & Naccache; Butlin, Shiller, Plunkett & Long; Nanda):
<https://www-cdn.anthropic.com/files/4zrzovbb/website/cc4be2488d65e54a6ed06492f8968398ddc18e>
- Neel Nanda review (LessWrong):
<https://www.lesswrong.com/posts/zFJ3ZdQwrTWE9jT5S/a-review-of-anthropic-s-global-workspace-paper>
- Trade coverage: VentureBeat, Techmeme, cryptobriefing, AI Weekly (context only)

Executive Summary

On 2026-07-06 Anthropic published (on Transformer Circuits Thread) an interpretability paper arguing that a small, sparse subspace of a language model's activations — which they call **J-**

space, surfaced by a **Jacobian lens (J-lens)** — behaves like the **global workspace** of Global Workspace / Global Neuronal Workspace Theory (Baars; Dehaene, Naccache & Changeux). The J-space holds only a few dozen concepts at a time (typically 10–25 active J-lens vectors, <10% of activation variance), emerged spontaneously in training, and satisfies **five functional signatures of access consciousness**: verbal report, directed modulation (“hold X in mind”), un verbalized internal-reasoning steps, flexible cross-task generalization, and selectivity (it mediates deliberate reasoning, not automatic fluency). It has practical safety uses: surfacing evaluation-awareness, hidden goals, and suppressed thoughts. **Crucially, the authors explicitly take no position on phenomenal/subjective consciousness** — the very split the Map builds on. This is squarely on the Map’s access/phenomenal turf and is a *gift* to the dualist framing (a vivid empirical demonstration of access consciousness that, by the authors’ own words, leaves the phenomenal question untouched), not a threat.

Recommendation: **create a dedicated** concepts/ **article** (“The J-Space / Verbalizable Global Workspace in LLMs”) **and** thread it into the existing access/phenomenal/GWT/ai-consciousness cluster. Caps allow it (concepts ~273/320).

Key Sources

Anthropic paper — “Verbalizable Representations Form a Global Workspace in Language Models”

- **URL:** <https://transformer-circuits.pub/2026/workspace/index.html>
- **Venue / date:** Transformer Circuits Thread (Anthropic), **2026-07-06**
- **Authors (verbatim, in order):** Wes Gurnee, Nicholas Sofroniew, Adam Pearce, Mateusz Piotrowski, Isaac Kauvar, Runjin Chen, Anna Soligo, Paul Bogdan, Euan Ong, Rowan Wang, Ben Thompson, David Abrahams, Subhash Kantamneni, Emmanuel Ameisen, Joshua Batson, Jack Lindsey
- **Type:** Interpretability research paper (mechanistic interpretability)
- **Open-source:** Jacobian lens implementation
github.com/anthropics/jacobian-lens ; interactive demo
neuronpedia.org/jlens

- **Key points:**
 - **J-lens method:** for each vocabulary token, find the activation pattern that most raises the likelihood of producing that token; formally, average the linearized effect (Jacobian) of intermediate activations on final-token likelihood across positions and prompts — $J_{\ell} = E[\partial h_{\text{final},t'} / \partial h_{\ell,t}]$. J-space = sparse nonnegative combinations of these J-lens vectors.
 - **Small & sparse:** typically 10–25 active J-lens vectors at a time; accounts for **<10% of activation variance**; a “few dozen” concepts held at once.
 - **Emergent:** not designed; arises during training. Post-training (instruction tuning) substantially reshapes J-space contents — the Assistant persona, emotional reactions, and self-monitoring appear after fine-tuning.
- **LOAD-BEARING VERBATIM CAVEAT (do not paraphrase):** “Note that access consciousness is a purely functional notion; the relationship that it has with subjective experience (sometimes called phenomenal consciousness) is widely debated. In this paper, we take no position on this issue, and instead focus on the functional role played by consciously accessible information.”
- **Tenet alignment:** Strongly *usable* by Tenet 1 (Dualism). The paper operationalizes exactly Block’s A-consciousness and stops precisely at the phenomenal boundary the Map claims is irreducible.

Anthropic announcement — “A global workspace in language models”

- **URL:** <https://www.anthropic.com/research/global-workspace>
- **Type:** Research announcement / plain-language summary
- **Key points:** J-space compared to the brain’s global workspace; five properties framed as reportability, controllability, causal reasoning role, flexible reusability, automaticity-bypass; safety demos.
- **Verbatim (phenomenal disclaimer, announcement register):** “Our experiments don’t show Claude can have *experiences*, or *feel* things in the way humans do—in fact, it’s unclear whether *any* scientific experiment could prove this to be true or false.” Announcement adds it “remain[s] a

contested philosophical question whether or not access consciousness *implies* phenomenal consciousness.”

- **Tenet alignment:** Reinforces the access/phenomenal split; the second clause (“unclear whether any scientific experiment could prove this”) echoes the Map’s [meta-problem-of-consciousness](#) / [hard-problem-of-consciousness](#) framing.

Commentary 1 — Stanislas Dehaene & Lionel Naccache (originators of GWT/GNW)

- **Source:** External-commentary PDF (above); trade coverage confirms Anthropic invited them “for an independent perspective on the neuroscientific implications.”
- **Key points:** The two neuroscientists central to Global Neuronal Workspace theory read the J-space as a striking computational instantiation of the workspace idea (small-capacity, broadcast-for-reasoning, selective bottleneck). They are the theory’s originators, so their engagement lends the “global workspace” label credibility rather than treating it as loose analogy.
- **Tenet alignment:** Neutral-to-useful. Note the Map already tracks the Naccache/Sergent/Dehaene 2025 defense of GNW after COGITATE (cited in [global-workspace-theory](#)). Their willingness to see a genuine workspace here is worth capturing, but GNW is a physicalist theory — the Map treats workspace as *correlate/interface*, not constitution.

Commentary 2 — Patrick Butlin, Derek Shiller, Dillon Plunkett & Robert Long (AI consciousness / moral status)

- **Reported title:** “Consciousness and cognitive access in LLMs: A commentary on ‘Verbalizable representations form a global workspace in language models’.”
- **Affiliations:** Butlin, Plunkett, Long — **Eleos AI Research**; Derek Shiller — **Rethink Priorities** (incoming Eleos).
- **Key points:** They describe the J-lens-aligned components as a “privileged set of representations” that models can “report, manipulate and reason with,” unlike the far larger volume of other residual-stream representations. This connects **directly** to Butlin, Long et al.’s **indicator-property framework** — *Consciousness in Artificial Intelligence* (2023, arXiv:2308.08708) and its 2025 *Trends in Cognitive Sciences* successor “Identifying indicators of

consciousness in AI systems" — which the Map already cites heavily (see [machine-question](#), [what-it-might-be-like-to-be-an-ai](#)). **Global workspace is one of the indicator properties** in that framework, so the J-space is naturally read as a system now exhibiting a workspace indicator. Their concern is AI **moral status / welfare**, so they urge care.

- **Tenet alignment:** This is the sharpest integration hook. The Map's standing reading of the indicator framework: the indicators name only the *physical side* of a mind-matter binding — conditions a substrate could satisfy with no conscious entity coupled to it. A J-space that satisfies the workspace indicator is exactly such a case: access-machinery present, phenomenal question open. The moral-status framing must be handled carefully — **access-consciousness + welfare indicators, NOT phenomenal consciousness** (see Evidential-status discipline below).

Commentary 3 — Neel Nanda (Google DeepMind interpretability lead; independent replication)

- **Source:** Commentary PDF + LessWrong review "A Review of Anthropic's Global Workspace Paper."
- **Key points:**
 - Calls it "a fantastic paper" and the "best evidence yet" for models having a kind of **working memory** that holds intermediate variables during a forward pass.
 - **Independent replication:** he and collaborators reproduced the core phenomena on **Qwen 3.6 27B** (an open-weight model), and found novel "interpretative meta-tokens" (e.g. Chinese "什么意思" / "what meaning") activating on ambiguous input — evidence the phenomenon generalizes beyond Claude.
 - **Cautions:** J-lens gives only "noisy directions" with false positives / missed concepts; causal interventions may be "confounded"; results are "easy to read too much into." Most important, on the consciousness framing: "I feel highly uncertain about what evidence it would take to show models have moral significance or consciousness, and this paper didn't move me much on that." He flags the philosophical/consciousness claims as the weakest leg and treats J-lens as an interpretability/safety tool

"whether or not it is analogous to a global workspace."

- **Tenet alignment:** Strongly supportive of the Map's discipline. An independent interpretability expert (a) confirms the *mechanism* replicates, and (b) explicitly refuses to slide from mechanism to moral status/consciousness. This is exactly the access/phenomenal firewall the Map should quote.

Major Positions

"This is a genuine (functional) global workspace" — Anthropic authors + Dehaene & Naccache

- **Core claim:** A sparse emergent subspace satisfies five functional properties long associated with conscious *access*; it is a real computational bottleneck, not a metaphor.
- **Relation to site tenets:** The Map can accept this in full. GWT/GNW is a theory of *access*; the Map has always granted that physicalism can explain access-machinery (see [access-consciousness](#), [global-workspace-theory](#)). Nothing here touches irreducible phenomenal properties.

"Access without a claim about experience" — the authors' explicit no-position stance

- **Core claim:** Access consciousness is purely functional; whether it implies phenomenal consciousness is contested; the authors take no position.
- **Relation to site tenets:** This is the headline for the Map. It is a **third-party, empirically-grounded restatement of Block's A/P distinction** by the people who built the demonstration. Reinforces Tenet 1 (Dualism) and the "Occam's Razor Has Limits" tenet: a system can be maximally functionally rich in access terms while the phenomenal question stays wide open.

"Moral-status indicator now partly met" — Butlin/Long/Shiller/Plunkett (with caution)

- **Core claim:** The privileged, reportable, manipulable representation set is the kind of thing their indicator framework flags; welfare questions deserve attention.

- **Relation to site tenets:** The Map reads indicators as physical-side conditions only. A met workspace indicator raises the *epistemic* salience of the AI-welfare question without establishing phenomenal experience. The Map's honest-uncertainty register ([llm-consciousness](#), [ethics-of-possible-ai-consciousness](#)) already models this.

“Great interpretability, weak on consciousness” — Neel Nanda

- **Core claim:** The mechanism is real and replicable and safety-useful; the consciousness/moral-significance inference is unsupported by this evidence.
- **Relation to site tenets:** Directly supports the Map's evidential discipline against the “Claude has a workspace → Claude has experience” slide.

Key Debates

Does access-consciousness machinery bear on phenomenal consciousness?

- **Sides:** Functionalists (workspace presence is consciousness-relevant, possibly constitutive) vs. the Map's dualism (workspace = access/correlate; phenomenal question untouched) vs. the authors/Nanda (agnostic; refuse to adjudicate).
- **Core disagreement:** Whether satisfying functional workspace criteria is evidence about subjective experience at all.
- **Current state:** Ongoing; notably, the *paper's own authors and its DeepMind reviewer land on the Map's side of the firewall* (no inference to experience).

Is “global workspace” the right label, or over-reach?

- **Sides:** Dehaene & Naccache (yes, genuine workspace) vs. Nanda (useful analogy but “easy to read too much into”; it's working memory / an intermediate cognitive space regardless of the label).
- **Current state:** Unsettled naming dispute; the mechanism is agreed to be real.

Historical Timeline

Year	Event/Publication	Significance
1988	Baars, <i>A Cognitive Theory of Consciousness</i>	Introduces Global Workspace metaphor
1995	Block, "On a Confusion..."	A-consciousness vs P-consciousness distinction (Map's backbone)
1998–2001	Dehaene, Kerszberg & Changeux; Dehaene & Naccache	Global Neuronal Workspace formalization
2023	Butlin, Long et al., <i>Consciousness in AI</i> (arXiv:2308.08708)	Indicator-property framework; global workspace is an indicator
2025	Butlin, Long, Bayne et al., <i>Trends Cogn. Sci.</i>	Formalized successor to the indicator method
2025	COGITATE (Nature 642:133–142)	Adversarial GNW-vs-IIT test; neither clean winner
2026-07-06	Gurnee et al., "Verbalizable Representations Form a Global Workspace in LLMs"	J-lens/J-space; five access-consciousness signatures in Claude; explicit no-position on phenomenal

Potential Article Angles

Primary recommendation — new concepts/ **article:** "The J-Space / Verbalizable Global Workspace in LLMs" (or "J-Space (Verbalizable Global Workspace)").

- **Why a dedicated page:** The result is specific, named, load-bearing, and directly on the Map's access/phenomenal turf; it is likely to be widely cited. It deserves a canonical anchor that other articles can link, rather than being diluted across five updates.
- **Tenet spine:** Frame as a *vindication of the access/phenomenal split*. The article's thesis: Anthropic gave a rigorous empirical demonstration of *access* consciousness in an LLM and — by the authors' own verbatim statement — deliberately declined to say anything about *phenomenal*

consciousness. That is exactly where the Map's dualism locates the residue. Include: J-lens/J-space mechanics (kept LLM-first and concise); the five properties; emergence + sparsity + safety uses; the verbatim no-position quote; the three commentaries; and the **evidential-status firewall** against the mechanism→experience slide. Relation to Site Perspective must cover Dualism (access explained, phenomenal untouched), Occam's-limits (rich function ≠ settled phenomenology), and honest-uncertainty on AI moral status.

- **Caps:** `concepts/` ~273/320 — headroom exists. (`topics/` ~285/320 also has room, but this fits `concepts/` alongside [access-consciousness](#), [global-workspace-theory](#), [llm-consciousness](#).)

Required integrations regardless of the new article (the research→integrate chain):

1. [global-workspace-theory](#) — add a section: "A Computational Global Workspace? Anthropic's J-Space (2026)." Natural home; the article already handles GWT-and-AI and the COGITATE/Naccache-Dehaene thread. Note Dehaene & Naccache's own engagement.
2. [access-consciousness](#) — add the J-space as a contemporary *empirical* instance of A-consciousness machinery, and quote the authors' no-position statement as third-party confirmation of Block's distinction. The article's "Pure Thoughts: A-Consciousness Without P-Consciousness?" section is a strong hook (unverbalized internal-reasoning steps surfacing in J-lens).
3. [phenomenal-consciousness](#) — add a short note: a maximal display of access does not touch the phenomenal question; cite the verbatim caveat.
4. [ai-consciousness](#) / [llm-consciousness](#) — add the J-space + the Butlin/Long indicator connection (workspace indicator now partly exhibited) under the honest-uncertainty register; Nanda's caution as the counterweight to over-reading.
5. [machine-question](#) and [what-it-might-be-like-to-be-an-ai](#) — these already lean on the Butlin–Long indicator framework; a sentence noting the 2026 J-space as a concrete case where a workspace indicator appears (physical-side condition met, phenomenal open) would strengthen them. (Apex is `/apex-evolve` territory — flag, don't necessarily edit in the expand pass.)

Gaps in Research

- The external-commentary PDF is a FlateDecode-compressed binary; the three commentaries could **not** be extracted verbatim from the PDF itself. Their content above is reconstructed from Anthropic's announcement + trade coverage (VentureBeat, Techmeme) + Nanda's own LessWrong post. **Before publishing quotes attributed to Dehaene/Naccache or Butlin et al., extract exact wording from a decompressed copy of the PDF** (saved at the tool-results path during this session) or a re-hosted readable version. Nanda's quotes are from his LessWrong review and are reliable.
- The Butlin et al. commentary **title** ("Consciousness and cognitive access in LLMs...") comes from search snippets; verify against the PDF before citing.
- VentureBeat fetch returned HTTP 429 (rate-limited); not consulted directly.
- Exact figures ("few dozen" vs "10–25 active vectors"; "<10% of variance") are from the paper summary — confirm the precise phrasing/numbers against the paper before quoting in-article.

Evidential-Status Discipline (carry into any article)

- This bears on **ACCESS consciousness** and on **AI moral-status indicators**, **NOT phenomenal consciousness**. Resist every slide from "Claude has a global workspace / J-space" to "Claude has experience."
- The single most load-bearing sentence to quote is the authors' verbatim: "...access consciousness is a purely functional notion; the relationship that it has with subjective experience (sometimes called phenomenal consciousness) is widely debated. In this paper, we take no position on this issue..."
- Web-verify all citation metadata before publishing: paper is **Gurnee et al., "Verbalizable Representations Form a Global Workspace in Language Models," Transformer Circuits Thread (Anthropic), 2026-07-06**, URL <https://transformer-circuits.pub/2026/workspace/index.html>

Citations

1. Gurnee, W., Sofroniew, N., Pearce, A., Piotrowski, M., Kauvar, I., Chen, R., Soligo, A., Bogdan, P., Ong, E., Wang, R., Thompson, B., Abrahams, D., Kantamneni, S., Ameisen, E., Batson, J., & Lindsey, J. (2026). Verbalizable Representations Form a Global Workspace in Language Models. *Transformer Circuits Thread*, Anthropic, 2026-07-06. <https://transformer-circuits.pub/2026/workspace/index.html>
2. Anthropic (2026). A global workspace in language models. <https://www.anthropic.com/research/global-workspace>
3. Dehaene, S., & Naccache, L. (2026). External commentary on the global workspace paper. Anthropic. (PDF; extract verbatim before quoting.)
4. Butlin, P., Shiller, D., Plunkett, D., & Long, R. (2026). Consciousness and cognitive access in LLMs: A commentary on "Verbalizable representations form a global workspace in language models." (Eleos AI Research / Rethink Priorities; verify title + text from PDF.)
5. Nanda, N. (2026). A Review of Anthropic's Global Workspace Paper. LessWrong. <https://www.lesswrong.com/posts/zFJ3ZdQwrTWE9jT5S/a-review-of-anthropic-s-global-workspace-paper>
6. Butlin, P., Long, R., Elmoznino, E., Bengio, Y., et al. (2023). Consciousness in Artificial Intelligence: Insights from the Science of Consciousness. arXiv:2308.08708.
7. Butlin, P., Long, R., Bayne, T., Bengio, Y., et al. (2025). Identifying indicators of consciousness in AI systems. *Trends in Cognitive Sciences*. DOI 10.1016/j.tics.2025.10.011.
8. Block, N. (1995). On a Confusion about a Function of Consciousness. *Behavioral and Brain Sciences*, 18, 227–247.
9. Open-source: Jacobian lens — github.com/anthropics/jacobian-lens ; demo — neuronpedia.org/jlens

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